#!/usr/bin/env python3

"""
SpiralCore Acoustic Air-Boot & Sound Wave Transceiver
Encodes 19-Bit UMT Tokens & 32-Byte Machine Images into Bell 2-of-6 Goertzel Audio Chimes.
Decodes audio back into UMT tokens with SHA-256 verification.
100% Pure Python + NumPy + Standard Library 'wave' & 'struct'.
"""

import sys
import os
import math
import struct
import wave
import hashlib
from typing import Tuple
import numpy as np

# Bell 2-of-6 Frequency Bank (Hz)
GOERTZEL_FREQS = [697, 770, 852, 941, 1209, 1336, 1477, 1633]

class AcousticAirBootTransceiver:
def __init__(self, sample_rate: int = 44100, tone_duration: float = 0.05):
self.sample_rate = sample_rate
self.tone_dur = tone_duration

def encode_umt_to_chime(self, umt_token: int, filename: str =
"umt_air_boot.wav") -> str:
"""

```
Converts a 19-bit UMT token (e.g. 0x0F91B) into a sequence of dual-tone audio chimes.
"""
hex_str = f"{umt_token:05X}"
audio_samples = []

for char in hex_str:
val = int(char, 16)
f1 = GOERTZEL_FREQS[val % 4]
f2 = GOERTZEL_FREQS[4 + (val // 4)]

t = np.linspace(0, self.tone_dur, int(self.sample_rate * self.tone_dur), False)
tone = 0.5 * (np.sin(2 * np.pi * f1 * t) + np.sin(2 * np.pi * f2 * t))

envelope = np.sin(np.pi * np.linspace(0, 1, len(t)))
tone = tone * envelope

audio_samples.extend(tone)
audio_samples.extend(np.zeros(int(self.sample_rate * 0.01)))

audio_arr = np.array(audio_samples)
audio_int16 = (audio_arr * 32767).astype(np.int16)

out_path = os.path.join(os.getcwd(), filename)
with wave.open(out_path, "wb") as wf:
wf.setnchannels(1)
wf.setsampwidth(2)
wf.setframerate(self.sample_rate)
wf.writeframes(audio_int16.tobytes())

return out_path
```

```
def decode_chime_to_umt(self, wav_path: str) -> Tuple[int, str]:
"""
Decodes a .wav audio file back into the 19-bit UMT token via FFT Goertzel frequency estimation.
"""
with wave.open(wav_path, "rb") as wf:
frames = wf.readframes(wf.getnframes())
data = np.frombuffer(frames, dtype=np.int16) / 32767.0
sr = wf.getframerate()

chunk_size = int(sr * self.tone_dur)
gap_size = int(sr * 0.01)
step = chunk_size + gap_size

hex_chars = []
for i in range(0, len(data), step):
chunk = data[i : i + chunk_size]
if len(chunk) < chunk_size // 2:
break

fft_vals = np.abs(np.fft.rfft(chunk))
freqs = np.fft.rfftfreq(len(chunk), 1.0 / sr)

low_band = [f for f in GOERTZEL_FREQS[:4]]
high_band = [f for f in GOERTZEL_FREQS[4:]]

f1_est = low_band[np.argmax([fft_vals[np.argmin(np.abs(freqs - f))] for f in low_band])]
f2_est = high_band[np.argmax([fft_vals[np.argmin(np.abs(freqs - f))] for f in high_band])]

f1_idx = GOERTZEL_FREQS[:4].index(f1_est)
```

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f2_idx = GOERTZEL_FREQS[4:].index(f2_est)

digit = f2_idx * 4 + f1_idx
hex_chars.append(f"{digit:X}")

reconstructed_hex = "".join(hex_chars)
umt_val = int(reconstructed_hex, 16) if reconstructed_hex else 0
sha256_hash = hashlib.sha256(reconstructed_hex.encode()).hexdigest()[:8]

return umt_val, sha256_hash

def run_acoustic_demo():
print("\033[96m\033[1m")
print("╔═══════════════════════════════════════════════════════════════════════════════════════════════════════════════╗")
print("║ SPIRALCORE ACOUSTIC AIR-BOOT & SOUND WAVE TRANSCEIVER ║")
print("║ Bell 2-of-6 Goertzel Audio Chimes + 19-Bit UMT Decoder ║")
print("╚═══════════════════════════════════════════════════════════════════════════════════════════════════════════════╝")
print("\033[0m")

transceiver = AcousticAirBootTransceiver()
test_umt = 0x0F91B # Synthesis Token Payload

print(f"1. Encoding UMT Token \033[96m0x{test_umt:05X}\033[0m to Goertzel audio chimes...")
wav_path = transceiver.encode_umt_to_chime(test_umt, "umt_air_boot.wav")
print(f" \033[92m✓ Audio chime generated successfully: {wav_path}\033[0m\n")

print(f"2. Air-Boot Decoding audio wave back to state payload...")
decoded_umt, sha_hash = transceiver.decode_chime_to_umt(wav_path)
```

```
print(f" Decoded UMT Token: \033[96m0x{decoded_umt:05X}\033[0m")
print(f" SHA-256 Parity: \033[95m{sha_hash}\033[0m")

if decoded_umt == test_umt:
print("\n\033[92m\033[1m✓ ACOUSTIC AIR-BOOT SUCCESSFUL! 100% BIT-EXACT MATCH OVER SOUND WAVES!\033[0m\n")
else:
print("\n\033[91m! Decoded payload parity mismatch.\033[0m\n")

if __name__ == "__main__":
run_acoustic_demo()
```

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